



A Bayesian analysis of topographic influences on the presence and severity of beech bark disease



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ABSTRACT

Forests of the Eastern United States and North Eastern Canada have been devastated by the onslaught of beech bark disease (BBD). The cultivation and management of resistant trees is an important conservation issue. Disease resistance is an interplay between tree genetics and environmental factors. To further understand resistance to BBD, this study investigates the contribution of topographic factors to both the occurrence and the severity of beech bark disease through the examination of small-scale patterns of disease. Approximately two hundred trees of the species *Fagus grandifolia* (American beech) located on an eighty-five-acre plot in south-western Vermont were monitored over five years for the presence and severity of BBD. Trees were examined yearly, during which time diameter at breast height (DBH) was recorded as well as a measure of disease severity. Topographical factors (aspect, slope, curvature, elevation, and aspect • slope) were assessed as potential predictors of disease presence and severity. Two distance measures of disease presence in nearby trees were also included to control for the spatial spread of the disease. Akaike's Information Criterion (AIC) was used to determine which subset of parameters yielded the best binary and ordinal logistic models of disease presence and rank. Bayesian analysis was used to determine the joint posterior distribution for model parameters. Slope was a both significant factors in determining disease presence. Slope • aspect was significant in determining both the presence and disease severity. DBH, time, and weighted distance to diseased trees were strong indicators of both disease presence and severity. Curvature and elevation were not significant factors. Predictive models based on identified local topographical parameters can identify environmental factors conducive to resistance which can be utilized to inform and designate regional beech restoration sites. Such models can also identify potential candidate trees for resistance screening and genetic profiling. With the more recent availability of higher resolution USGS topographic maps for the northeastern United States, it is easier to determine and evaluate ideal resistant beech habitat. From these maps we can predict regions with healthier beech trees and suggest potential ecologically favorable restoration sites only using slope and aspect.

1. Introduction

1.1. History of the American beech and beech bark disease

There are eleven species of beech worldwide, yet the American beech (*Fagus grandifolia*) is the only beech native to North America, having carved out a critical ecological niche in the forests of Southeastern Canada and the Eastern United States (Huntley et al. 1989). Although American beech is overshadowed by its European relatives in terms of economic value, the species has high ecological

value. This late successional tree is one of the most prominent hardwood tree species in North America. Similar to their European counterparts, American beech trees, growing to approximately one hundred feet and living up to 400 years, provide high quality habitat and have positive impacts on both soil chemistry and richness (Boerner and Koslowsky 1989, Lovett et al. 2002). American beech is notably one of only three mast tree species in its native regions and may be the only mast producer in a given system. Beechnuts provide a rich fat and protein resource to many species including deer, fox, bear, ruffed grouse, blue jays and wild boar (Tubbs and Houston 1990, Sabrina

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et al., 2009, Morris et al., 2004), and mast availability can also influence those animals that depend on small rodents as their primary diet source (Lovett et al. 2006).

Despite the extensive logging and clearing that mark the post-colonial era in the United States, beech were able to reestablish relatively quickly in second-growth forests. In the early part of the twentieth century, the pathogens responsible for Beech Bark Disease (BBD) in Europe were inadvertently introduced to the Americas through the importation of ornamental beech. Starting in Halifax, Nova Scotia, likely in 1890, the disease effectively spread south to the Carolinas and west through Quebec and Ontario and to Michigan in the United States by 2015 (Cale et al. 2017) showing no signs of not eventually reaching the full ecological range of the American beech that stretches to northern Florida and eastern Texas (Stephanson and Coe 2017). American beech has steadily declined in abundance and vigor since the introduction of the disease-causing agents (Mize and Lea 1979, Gavin and Peart 1993, Houston 1994).

It is well-documented that invasions of exotic pests pose significant threats to forested ecosystems (Leibold 1995, Roy et al. 2014) and that the effects may be compounded by stochastic events such as forest fires as well as by climate change (van Lierop et al. 2015, Langor and Alfaro 2016, Stephanson and Coe 2017). The accidental or intentional release of pests by humans have exposed forests of the Northeastern United States to a great number of forest disturbance agents such as the gypsy moth, hemlock woolly adelgid, chestnut blight, and the beech bark disease complex (Morin et al. 2007, Paradis et al. 2008, Dukes et al. 2009). The impact of these invasions on forest structure, management, and community dynamics has been significant (Lovett et al. 2006). BBD has been and continues to be one of the most devastating tree epidemics affecting northeastern forests (Morin et al. 2005, Lovett et al. 2006).

BBD begins with an initial attack by the beech scale (*Cryptococcus fagisuga* Lindinger,

1936; Hemiptera: Eriococcidae) followed by subsequent infection by one or more

varieties of Ascomycete fungus, that may include *Nectria distissima* (Castlebury et al., 2006), *Neonectria faginata* (Roy and Nolet 2018), and *Nectria galligena* Bres. (Faison and Houston, 2004). *Nectria distissima* is native to Europe, whereas *Nectria faginata* is native to North America (Castlebury et al.). The scale insect bores minute holes selectively into susceptible beech bark which provides suitable habitat for the scale insect reproductive cycle. Susceptibility to BBD may be linked to overall tree health, including bark nitrogen and phosphate stores (Latty et al. 2003, Wargo 1988, Cale et al. 2017), stone cell layer structure (Lonsdale and Wainhouse, 1987) and/or underlying genetic factors (Seitler and Coe, 2015; Mason et al. 2013, Krabel and Petercord, 2000, reviewed in Stephanson and Coe 2017). The overwintering white wax that protects scale insect larvae is a clear and notable sign of BBD. Fungi enter the tree through the stylet formed holes upon departure of the scale insect. It is the fungal infection that leads to tree death as the function of storage and vascular cells is disrupted (Ehrlich 1934). Documented resistance is at the level of the scale insect but there is little natural genetic resistance to BBD in the American beech population, as there has been no coevolution of pathogen and host as has occurred in Europe (Houston 1994, Coe 2010). The infection in a typical stand ranges from 80 to 90% and estimates of true resistance are quite low. It has been hypothesized that only 1–3% of the American beech are resistant to the scale insect (Houston 1983, Kasson and Livingston 2012). To compound the severity of the issue, the pathogens have not yet reached the full beech range and have shown no signs of tapering off (Morin et al. 2005, Cale et al., 2017). There is evidence in Maine that forests can even be reinfected (Kasson and Livingston 2012).

North American forests through which BBD has passed experience changes at many levels. Shigo (1972) succinctly described three main phases of BBD: the advancing front, the killing front, and the aftermath stage. The advancing front consists of the beech scale insect with no or minimal sign of *Nectria* fungus. Trees in this stage are healthy and show

few signs of stress. The killing front occurs when *C. fagisuga* and one of the *Nectria* species inhabit the same forest. Beech trees at this stage show signs of severe stress: many are cracked and cankered, some are dead, while others demonstrate resistant to the disease. The aftermath stage is generally comprised of trees with all levels of disease, from dead individuals and severely infected trees (BBD rank of 3 or 4, further defined in Methods) to resistant stands. Resistance has been validated in field studies (Houston 1982). A combination of genetic predisposition and environmental impacts is hypothesized to confer the resistance documented in subsets of the American beech throughout its ecological range (reviewed in Stephanson and Coe 2017).

The physiology and phytopathology of BBD have been well documented and characterized throughout the United States and Canada (Ehrlich 1934, Griffin et al. 2003). BBD has been shown to affect all aspects of forest communities from stand structure to wildlife health (Foster et al. 1998, Latty et al. 2003, Faison and Houston 2004, Storer et al. 2005). The study of BBD spans nearly a century of research that has characterized the disease agents. As we have come to appreciate and more fully understand the distribution of the disease, efforts in the last decade have begun to focus on potential management practices that must be used to maintain American beech populations (Evans et al. 2005). Many of the relationships between BBD and ecosystems and overall ecosystem health remain unclear (Faison and Houston 2004). More recent attention has turned to the impacts of climate change that may influence and compound the effect on beech tree populations of BBD (Stephanson and Coe 2017).

1.2. Drivers of BBD resistance

Beech tree resistance to BBD has perplexed researchers for more than 50 years, but no definitive answers have yet been formulated to predict or explain BBD resistance. A research priority formulated from the 2004 Saranac Lake BBD Symposium was to determine the genetic mechanisms of resistance and tolerance of American beech to the BBD complex (Koch and Carey 2004). Limited but promising progress has been made through the use of proteomics and microarrays (Mason et al. 2013, Seitler and Coe 2015). Discernable changes in protein expression patterns have been documented between bark proteomes from susceptible and resistant trees (Mason et al. 2013). Microarray analyses have shown preliminary changes in three genes, including *chitinase* (Seitler and Coe 2015). These findings may provide a foundation for exploring the mechanism of resistance and perhaps allow for regional genotype resistance screening.

Concerted efforts by the USDA Forest Service Northern Research Station have pioneered a resistant beech breeding program (United States Department of Agriculture Forest Service 2010) which aims to eventually provide beech stock for regional plantings (Koch et al. 2012). This work is complemented by management strategies targeted at the specific stages of infestation (McCullough et al. 2005). Both efforts are abetted by an understanding of the relationship between environmental factors and susceptibility to BBD. Environmental variables such as slope and aspect have been linked to the severity and tolerance of BBD, but few fine-scale studies have been done to determine specifically how environmental variables relate to BBD susceptibility (Ehrlich 1934, Munck and Manion 2006). The development and modeling of intensively monitored small plots aids the study of how environmental conditions relate to BBD resistance (Evans et al. 2005) which can ensure optimal success in beech reintroduction efforts and also improve the study of BBD resistance by providing a prior estimate of the likelihood of a tree being diseased.

1.3. Bayesian modeling approach

Due to the high number of variables and their interactions, we chose Bayesian analysis for our modeling approach. Bayesian modeling has been used to examine disease transmission and progression in

ecosystems (Farnsworth et al. 2006), and the dynamics of various ecological processes (Wikle 2003). Ellison (2004) provides a detailed description of the advantages and uses of Bayesian analyses in ecological modeling. Both Hobbs and Hooten (2015) and Wade (2000) offer convincing arguments for the advantages of Bayesian analysis in ecology and conservation biology. An example of the applicability of a Bayesian approach to disease ecology is laid out in Panassiti et al. (2015), where Bayesian analysis was used to quantify the effects of environmental factors on the spread of the disease *bois noir* in a variety of grapevine species. We follow a similar approach in assessing the relevance and impact of environmental factors in relation to BBD resistance.

1.4. Hypotheses and purpose of study

In light of recent arguments asking for the consideration of “local differentiation” in management planning in the context of broader environmental issues such as climate change (Kramer et al., 2010), this study aims to characterize the local environmental profile that supports resistance to BBD in native beech. We hypothesize that the following environmental parameters impact both the likelihood and severity of BBD: slope, aspect, curvature and elevation. Slope has an impact on water availability, which is an indirect contributor to resistance to scale infection (Lonsdale, 1980). Hu et al. (2018) demonstrated that beech on slopes with a southern aspect receive more light which may increase beech resistance. Curvature is a measure of how rapidly slope and aspect change and, according to Latty et al. (2003), it can affect water availability and nutrient loads within soil. Lastly, elevation has been shown to have a significant effect on BBD by Ehrlich (1934), Griffin et al. (2003), and Wiggins et al. (2004).

In order to better estimate the impact of topographical variables, we also control for other hypothesized correlates of BBD. Diameter at breast height (DBH), time since start of study, and distance to the nearest diseased tree should all be strongly correlated with disease occurrence and extent. DBH was measured at the beginning of the study and is correlated with a tree's age at that time. DBH and time both control for the fact that older trees have had more time to be exposed to the disease and are therefore more likely to be infected. Distance accounts for the spatial dynamics of disease spread. Up to 99% of dispersing young scale insects migrate less than 10 m from their host tree (Wainhouse 1980). This could likely lead to clusters of infected trees. Further, the fungal vectors of BBD are often related to geographic variables and the proximity of diseased trees (Houston 1993).

By assessing ecological drivers of BBD, this study will impact restoration practices regionally and enable practitioners to target restoration projects towards locations likely to survive. It will give us a greater ability to identify trees with increased genetic resistance, and it will contribute to our understanding of generalizable trends that are relevant on a global scale in light of increased susceptibility to BBD.

The objective of this work is to examine the role of multiple environmental players (slope, aspect, slope • aspect, curvature) as potential drivers of BBD infection and severity to better evaluate the propensity of the disease itself but to tactically inform restoration efforts by predicting ecologically advantaged focus sites. Beech bark disease has ravaged the forested landscape of nearly the full range of beech in the United States. Ongoing restoration efforts that focus on genetic diversity and the development of more resistant stock can utilize this work regionally and our specific Bayesian approach can be modeled and modified for widespread applications to both BBD and other host-pathogen systems.

2. Methods

2.1. Site description

The Lewis Deane Nature Preserve (LDNP) is an ecologically diverse,

85-acre, forested parcel located at the northern end of the Taconic Mountains. LDNP straddles the towns of Poultney and Wells in Rutland County, Vermont and is host to more than eight plant communities including: hemlock forest, hemlock - northern hardwood forest, northern hardwood forest, white pine - northern hardwood forest, mesic red oak - northern hardwood forest, dry oak - hickory - hophornbeam forest, dry oak woodland, and a temperate acidic outcrop community. LDNP lies on the eastern shoulder of Saint Catherine Mountain, a north-south trending ridge. The site is characterized by slate bedrock, mainly from the Lake Saint Catherine formation. It has relatively steep slopes with predominantly eastern aspects. The steep slopes are broken by wide terraces that accumulate pooled water and form wetland conditions. Personal communication with landowners and aerial photographs have established that LDNP hosted sheep grazing at the turn of the twentieth century and cattle grazing later in the 1900s. By 1940, no livestock remained in the area, and forest reclaimed the pastureland. Selective logging practices were carried out in 1980 after which no extraction or grazing has occurred. At the time of this study, the American beech trees at LDNP had an average DBH of 17 cm. The DBH distribution showed that the stand is relatively young with some older trees that were likely spared in the selective logging process.

2.2. Environmental data collection

All American beech trees with DBH > 5 cm were identified and tagged with aluminum-zinc numbered markers (n = 204). Each tree was surveyed using a Topcon GTS-212 total station. Survey methods followed standard back-sight fore-shot techniques and shot to the center of each tagged American beech tree. Survey results were entered into Autodesk and a coordinate file was created. The coordinate file was then entered into ArcGIS Desktop 10.5 × (ESRI). A GIS shapefile was created representing each tagged American beech tree.

A hydro-flattened DEM derived from 0.7-meter LIDAR data was obtained from the Vermont Center for Geographic Information (2019). The DEM and the shapefile containing beech tree locations were imported into ArcMap 10.5. The Spatial Analyst extension was used to calculate slope, aspect, and curvature from the DEM. The Extract Multi Values to Points tool was used to join values from these layers, including the DEM, to each tree point feature. Distances between trees were calculated using Euclidean distance derived from UTM coordinates. The attribute table of the tree shapefile and the distance matrix were then exported for statistical analysis (see Section 2.5).

2.3. Monitoring of beech bark disease

Each of the American beech trees identified at LDNP was monitored over the course of five years. The rank assignment used the standard five-level system established by Griffin et al. 2003. This ranking system (quoted verbatim), is as follows:

- 1 = Very little or no sign of either causal agent (*Cryptococcus* or *Nectria*).
- 2 = *Cryptococcus* present; bark beginning to crack; tree still shows vigor. Canopy at least 75% intact.
- 3 = Bark heavily cracked; significant cankering from *Nectria* colonies; some crown damage or limb loss. Canopy 25–75% intact.
- 4 = Bark severely cracked; large girdling cankers; significant crown loss or snag.
- 5 = Dead from BBD

Individual trees were ranked every year for the five-year study. Since disease resistance occurs at the *Cryptococcus* stage, leaf litter was removed at the base of the tree to assess preliminary infection as the scale insect often initiates bark penetration there. Disease rankings for each year were completed by one investigator and ground checked by an additional investigator. Trees that were inadvertently ranked dead one year (rank = 5) and reported alive in later years had their ranks

changed to 4 (alive but with a severe infection) prior to analysis.

2.4. Model development

We desired to model both the spatial and temporal propagation of the disease as well as the severity of the disease within infected trees. Towards this end, we used binary logistic regression to predict the probability of a tree being infected with BBD ($\vec{P}_{infection}$), and we used ordinal logistic regression to model the temporal progression of the disease by estimating the probability of the degree of infection being rank n or below ($\vec{P}_{rank}$) for each year of the study. Slope, elevation, curvature, and aspect were included as predictor variables and treated as observed without error. Because aspect is a circular variable (360° equals 0°), both $\sin(\text{aspect})$ and $\cos(\text{aspect})$ were considered as factors. We also included interaction terms between slope and aspect ($\text{slope} \cdot \sin(\text{aspect})$ and $\text{slope} \cdot \cos(\text{aspect})$) as we hypothesized that the impact of aspect may be mediated by slope. The method of splitting a circular variable into the sine and cosine of that variable is taken from [Kim and SenGupta \(2016\)](#).

To control for variability in disease pressure, we included variables for age and proximity to diseased trees. DBH was used as a proxy for a tree's age at the beginning of the study. This is a reliable proxy within the context of this study given that all trees are within one study site. Time since year one was used to represent the change in age over the course of the study. Distance to the nearest diseased tree (the straight-line distance in meters to the nearest tree with a rank of 2–5) and an inverse-distance weighted sum referred to as the *weighted distance* ($\sum \frac{1}{d}$)

where $\vec{d}$ is the vector of straight-line distances to every other measured tree with a rank of 2–5) were used to estimate the disease pressure from nearby infected trees. These variables were also assumed as observed without error.

Akaike's information criterion (AIC) was used in a stepwise elimination procedure to identify the best subset of hypothesized factors for both the ordinal and logistic models. A Bayesian analysis was run on the best subset of factors for each model.

The following equation for the probability of infection and rank was developed:

$$P(BBD) = \phi(\theta - \vec{\beta} \cdot \vec{X}) \quad (1)$$

where ϕ is the logistic link function, θ is the intercept term, $\vec{\beta}$ is a vector of model coefficients, and $\vec{X}$ is the matrix of identified model factors. Note that for the ordinal logistic model, four separate intercept terms are needed. Following [Hobbs and Hooten \(2015\)](#), a full model diagram for the ordinal model is given in [Fig. 1](#). Non-informative priors were used for all variables as previous regional research examining these specific environmental attributes was not available.

The full posterior distribution for each model is given by:

$$[\theta, \vec{\beta}] \propto [\text{rank} | \vec{\beta}, \theta, \vec{X}] \cdot [\vec{\beta}] \cdot [\theta] \quad (2)$$

where $[x]$ indicates the probability distribution for the random variable

x.

2.5. Model estimation, specification, validation, and convergence

Posterior distributions were estimated using a Markov Chain Monte Carlo algorithm with Gibbs sampling ([Gelfand and Smith 1990](#), [King et al. 2009](#)). 10 chains were conducted for the logistic model and 5 chains for the ordinal. The Gelman-Rubin R statistic ([Gelman and Rubin 1992](#)) was used to determine a sufficient number of iterations to achieve convergence. Chains of 2,200,000 iterations were run with the first 200,000 discarded for burn-in. The remaining 2,000,000 were thinned by a factor of 500, yielding a final chain of 4000 values drawn from the joint posterior distribution. Tuning parameters for proposals were adjusted so that acceptance rates were generally between 20% and 40%.

Model specification was tested by simulating the number of trees of a given infection status for each link in the Markov chain and comparing the actual number of trees of each rank to the distribution of counts from the simulation. To further assess the validity of the ordinal model, three sets of simulated data were created using Eq. (1) with predetermined coefficient values and environmental variables randomly drawn from the empirical distribution of the actual data set. The probabilities of each rank for each sample were then used in a Monte Carlo simulation to produce a set of simulated disease ranks. Posterior distributions were then estimated for each coefficient using the simulated data and compared to the known coefficient values.

Following [Panassiti et al. \(2015\)](#), goodness-of-fit for the logistic model was assessed by calculating the area under the receiver operating characteristics curve (AUROC). Model residuals for the logistic model were plotted according to the latitude-longitude position of the corresponding tree to assess spatial auto-correlation. Bayesian p-values were calculated for all coefficients and estimate the probability that a given coefficient assumes a value strictly greater or less than zero. All calculations were performed in R 3.6.0. The MASS package was used to calculate AIC values.

2.6. Estimation of marginal probability functions

To visualize the effects for all factors whose coefficients had a Bayesian p-value less than 0.05, marginal cumulative probabilities for each rank as a function of the given factor were estimated by setting all coefficients to their median values from the Monte Carlo Markov Chain and allowing the factor of interest to vary over its observed range while setting other factors to their median values for the data set. This allows us to estimate and plot the change in tree rank probabilities as a function of each significant factor.

3. Results

At the onset of the study, 207 American beech trees were ranked with 199 included in the analysis. Of these, 118 trees showing little to no sign of infection, while the remaining 81 trees were in various stages

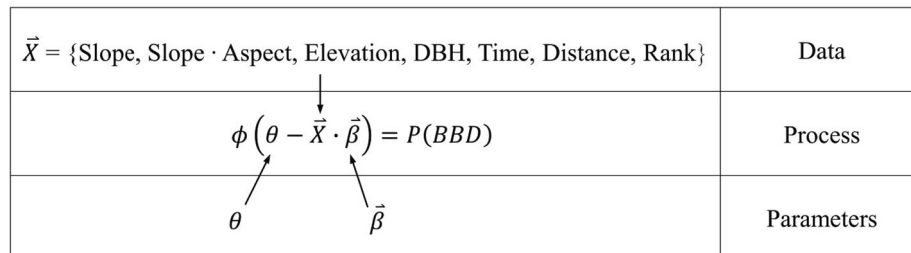


Fig. 1. A full model diagram for the analysis following [Hobbs and Hooten \(2015\)](#). ϕ is the logistic link function, θ is the intercept term (note that there are 4 in the ordinal model), and $\vec{\beta}$ is a vector of model coefficients.

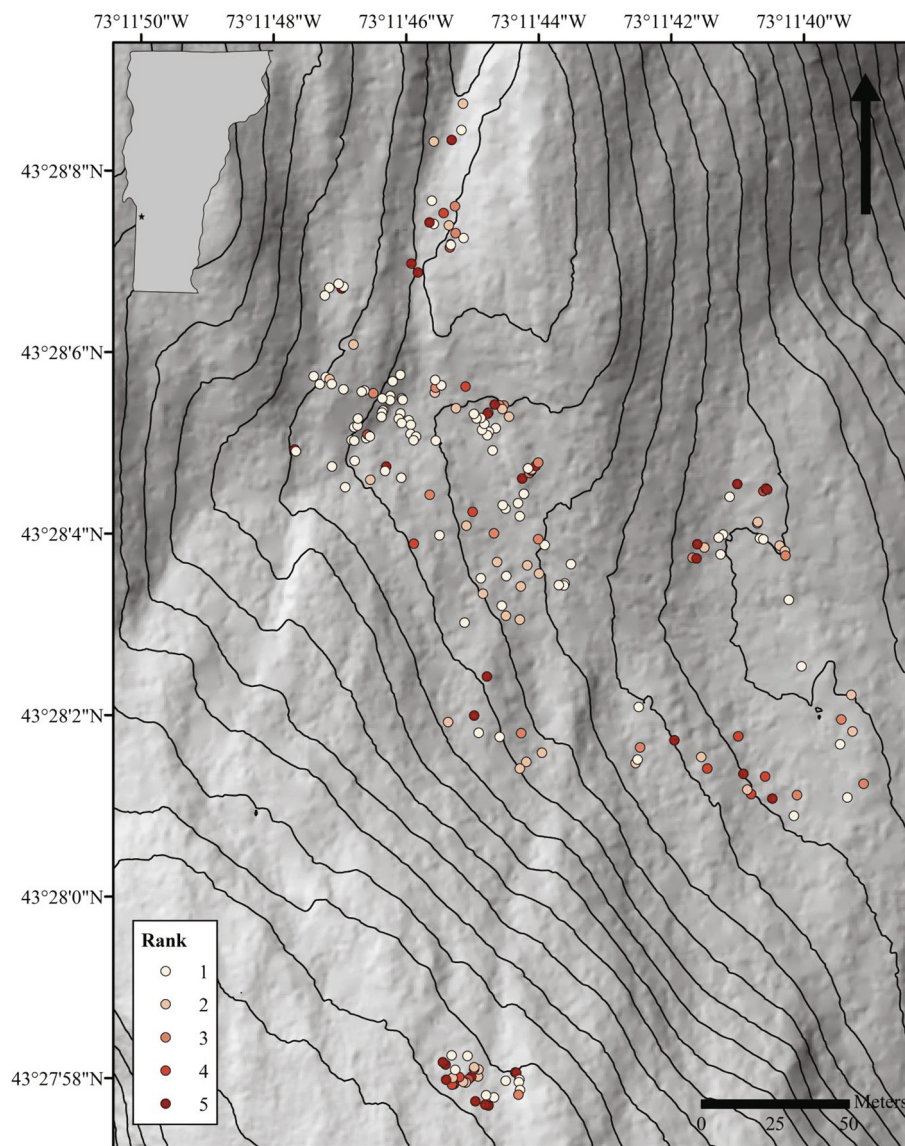


Fig. 2. Map of individual beech trees and color coded for disease ranking of BBD severity at the Lewis Deanne Preserve Study during the fifth year of observation. The site provides heterogeneity in the environmental parameters considered in the study.

of BBD. Over the course of the five years, the proportion of trees ranked 5 quadrupled with 14% of trees dead from clear signs of BBD by the last year of the study as compared to 3.5% in the first year. Additionally, 14% of those trees originally showing little to no sign of the disease (rank = 1) were ranked 2 or higher by year 5. Individual tree location and rankings for year five are shown in Fig. 2. Over the course of the study, data was gathered on 207 trees for each of the 5 years, although missing data caused the total number of samples to be 996.

Using AIC, we determined that the best logistic model was achieved by removing $\cos(\text{aspect})$, $\sin(\text{aspect})$, curvature, slope $\cdot \sin(\text{aspect})$, and the distance to the nearest diseased tree ($\Delta\text{AIC} = 33.137$ compared to the null model). The best ordinal model was achieved by removing the same parameters as well as slope ($\Delta\text{AIC} = 23.251$). Incorporating elevation improved AIC by only 1.030 for the logistic model and 1.139 for the ordinal. Previous work (Ehrlich 1934, Griffin et al. 2003, Wiggins et al. 2004) suggests that elevation is an important factor, and AIC analysis shows that it does improve the model and should therefore improve estimation of model coefficients. However, the effect of elevation is slight and mixed in our analysis (see Table 1). The AUROC for the logistic model suggested reasonably good fit (AUROC = 0.796).

There was no evidence of model misspecification. The simulated

disease rank frequencies did not differ significantly from the actual counts. Additionally, the joint posterior distributions estimated from the three sets of simulated data all contained the known coefficient values within a 90% confidence interval. The Gelman-Rubin statistic to test for chain convergence was less than 1.05 for all parameters in the logistic model and less than 1.06 for parameters in the ordinal model indicating that the Markov chains had converged. Spatial plots of model residuals for the logistic model show no evidence of spatial autocorrelation.

Median values for model coefficients, odds ratios, and respective Bayesian p-values are reported in Table 1. Posterior probability distributions for all variables for both models are reported in Fig. 3. Slope, and slope $\cdot \cos(\text{aspect})$ were significant variables in determining the probability of a tree being infected with BBD as were all disease pressure variables (DBH, time, and weighted distance). The same variables were significant in the ordinal regression model with the exception of slope which was not considered having been excluded by the AIC analysis. All variables that were significant were positively correlated with the probability that a tree would both be infected and have a greater infection level. These included slope (p-value for logistic < 0.001), $\cos(\text{aspect}) \cdot \text{slope}$ ($p_{\text{logistic}} < 0.001$, $p_{\text{ordinal}} < 0.001$), DBH

Table 1

Median values, odds ratios, and Bayesian p-values for model parameters from the Monte Carlo Markov Chain. A positive coefficient means a factor is correlated with increasing probability of disease. Every parameter that was significant was significant in both models, except for slope which was significant only in the logistic analysis. Elevation did not have a significant Bayesian p-value. Cos is the cosine of aspect. The per-unit change in odds ratio was calculated by taking the exponent of the median observed value.

	Slope	Slope · Cos	Elevation	DBH	Time	Weighted
Logistic						
Median	0.040	0.093	-0.0046	0.126	0.111	1.940
Odds Ratio	1.041	1.097	0.995	1.135	1.117	6.958
p value	< 0.001	< 0.001	0.17	< 0.001	< 0.05	< 0.001
Ordinal						
Median	NA	0.076	0.0028	0.120	0.099	1.806
Odds Ratio	NA	1.079	1.003	1.128	1.104	6.084
p value	NA	< 0.001	0.25	< 0.001	< 0.05	< 0.001

($p_{\text{logistic}} < 0.001$, $p_{\text{ordinal}} < 0.001$), time ($p_{\text{logistic}} < 0.05$, $p_{\text{ordinal}} < 0.05$), and weighted distance ($p_{\text{logistic}} < 0.001$, $p_{\text{ordinal}} < 0.001$). Elevation was not significant in either model which is consistent with the AIC analysis.

Based on the odds ratio, with all else being held constant, for each centimeter increase in DBH, the ratio of the probability of a tree being infected to the probability of it being healthy increases by 13.5% ($e^{0.126} = 1.135$). For each degree increase in slope, the ratio of the probability of a tree being infected to the probability of it being healthy increases by 4.1%. For each additional year that passes the odds ratio increases by 11.7%.

Cumulative probabilities for disease rank as a function of the significant factors are shown in Fig. 4a (logistic) and Fig. 4b (ordinal). Probabilities for individual ranks can be calculated as the difference in cumulative probabilities (e.g. probability of a tree having rank = 4 goes from approximately 2% at a DBH of 8 cm to approximately 14% at a DBH of 28 cm). Slope · aspect, DBH, and weighted distance have the largest effects on the probability of a tree having a higher disease ranking. Trees with a slope of approximately 14 degrees and a southern aspect (slope · cos(aspect) = -14) are almost twice as likely to be

healthy as trees with a slope of approximately 13 degrees and a northerly aspect (slope · cos(aspect) = 13). Trees with a DBH of 28 cm are three times more likely to be diseased than those with a DBH of 8 cm. Trees in close proximity to diseased trees (high weighted distance) have a probability of being healthy of less than 10%.

4. Discussion

Our study supports the role of slope and aspect as drivers of BBD infection and severity. Additionally, age and proximity to diseased trees were both strong predictors of disease presence and severity. Curvature and elevation were not shown to be significant factors in disease susceptibility. Determination of the topographic factors that influence BBD allows for the identification of ideal restoration sites and expedites future research focused on disease ecology.

Resistance to BBD is a highly complex phenomenon. There are many factors contributing to BBD presence and severity compounded by both environmental and genetic influences (Latty et al. 2003, Faison and Houston 2004, Munck and Manion 2006, Mason et al. 2013, Stephanson and Coe 2017). Physical attributes of the environment may

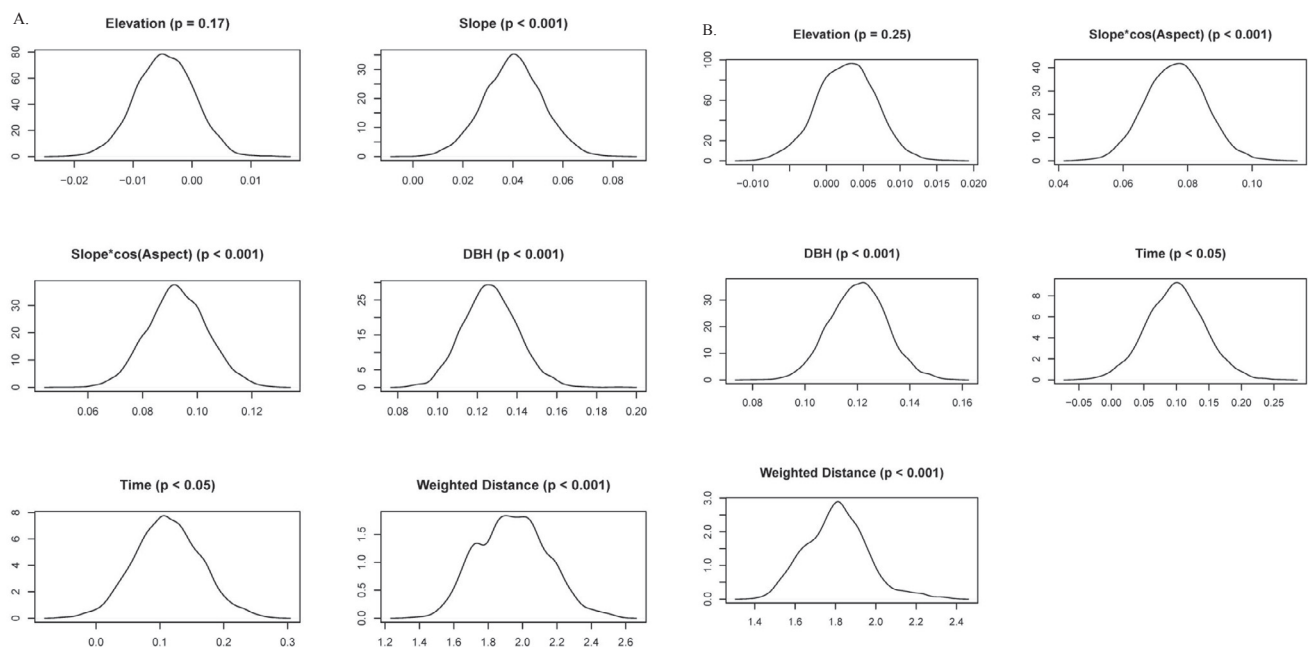


Fig. 3. Posterior probability distribution plots for model coefficients for the logistic (A) and ordinal (B) regression models. Distributions are estimated from the joint posterior distribution as represented by the Monte Carlo Markov Chain. The x axis shows the likely range for each model coefficient. The y axis shows probability density (making the area under the curve equal 1). Variable names and Bayesian p-values are given above each plot. The p-values report the proportion of the posterior distribution above (or below) zero and represent the likelihood that the variable has a significant effect on disease probabilities.

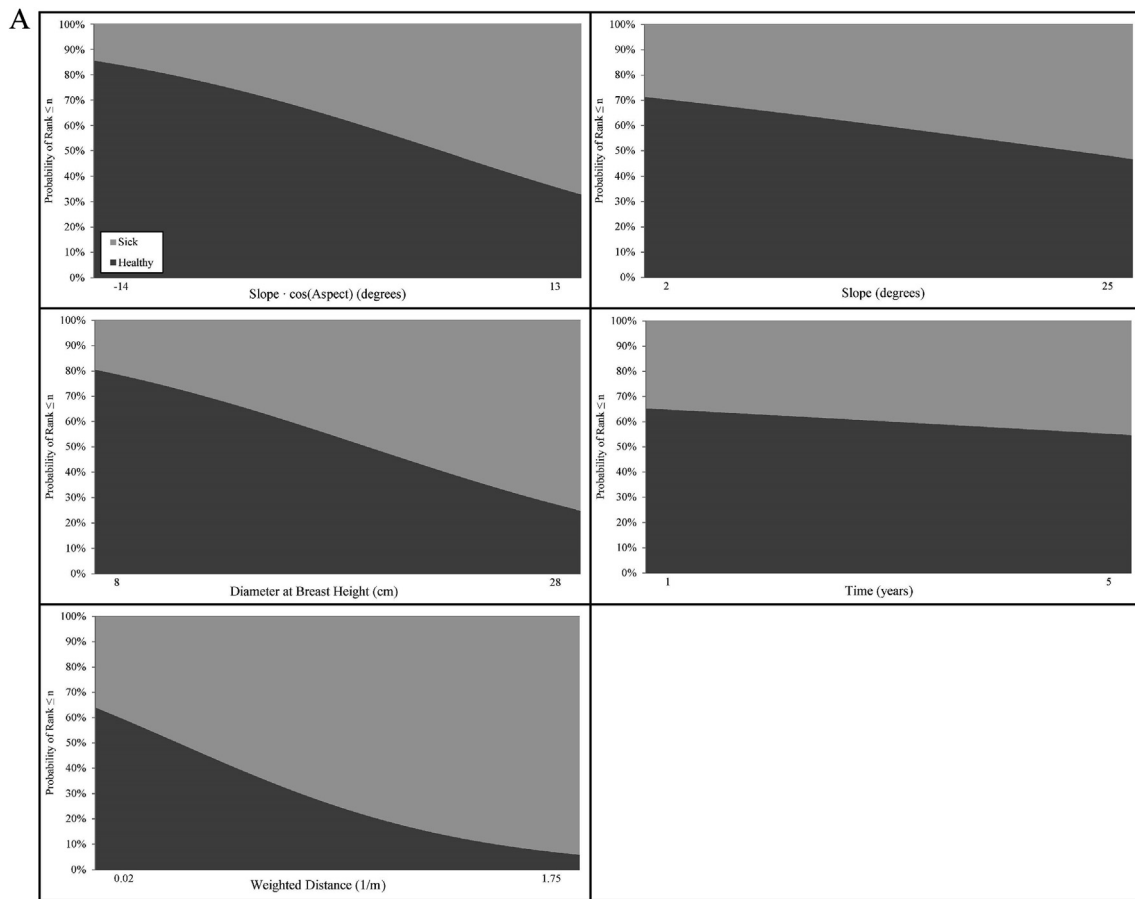


Fig. 4. Probability plots for disease presence or tree rank as a function of model parameters for the logistic (A) and ordinal (B) models. Plots are generated by setting all parameters except the independent variable at their median values and allowing the independent variable to vary over its observed range. Only significant parameters are displayed.

not only contribute to overall tree health and disease susceptibility or resistance (Lonsdale and Wainhouse 1987) but may also influence the ecology of the disease causal agents (Ehrlich 1934). The potential impact of environmental factors can aid in design and implementation of current restoration efforts. Here we have focused on physical attributes,

namely slope, aspect, curvature and elevation, and included a biological parameter (DBH) to determine if beech in a particular physical setting would have an increased likelihood of remaining healthy (or not) and if infected, how severe the infection might be.

Beech trees at southern aspect locations have a higher probability of

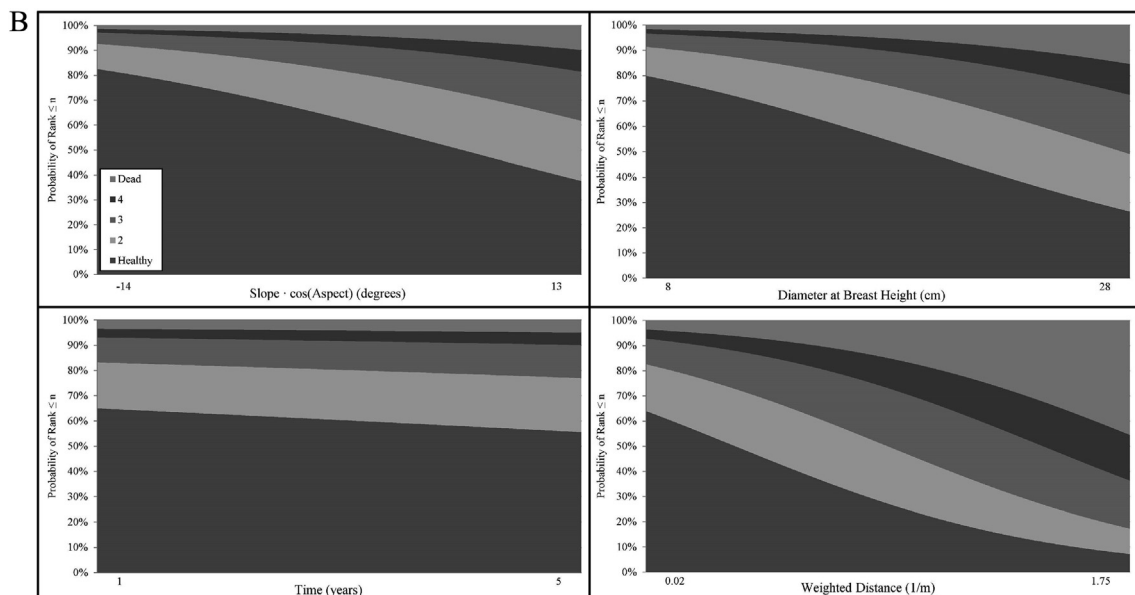


Fig. 4. (continued)

being scale insect free and healthier than those on northern facing slopes as slope • aspect was significant in both the logistic and ordinal models (Table 1) and the marginal probability plots show that trees with a negative slope • aspect (south-facing) are most likely to be healthy (Fig. 4), likely due to increased solar radiation as compared to other aspects (Hu et al. 2018). The warmer and drier conditions may preferentially influence overall beech health positively and negatively impact the scale insect microhabitat. Scale insects tend to colonize the north aspects (in relation to the tree itself) and prefer pockets of moderate moisture (Houston 1982, Spiller et al. 1990, Gullan and Kosztarab 1997, Boyce 2009). The significance of slope as an environmental factor is likely related to its impact on water availability which can be considered an indirect contributor to resistance to scale insect infestation (Lonsdale, 1980).

While curvature can influence local water availability and nutrient loads in soils (Latty et al. 2003), we did not see evidence of it being a factor. Similarly, elevation, despite having significant prior support for its impact on BBD (Ehrlich 1934, Griffin et al. 2003, and Wiggins et al. 2004), was not a significant factor at our site, most likely because we had relatively small variation in elevation (a total range of only 65 m).

Although environmental parameters such as aspect and slope influence BBD susceptibility, resistance is ultimately controlled by underlying genetic factors. Work with genetic crosses of non-infected parent beech revealed that “repeatabilities and family degree of genetic determination of beech scale resistance is sufficient to realize genetic gain” (Koch et al., 2010). This work intends to complement breeding and selection studies ongoing for beech restoration (United States Department of Agriculture Forest Service 2010, Koch and Carey 2014) by systematically evaluating individual environmental parameters to determine locations that are less susceptible to the scale insect. Additionally, this study will complement the continued development of management plans by Vermont Fish and Wildlife (Hamelin 2011) to increase mast in BBD affected forests and will also provide a backdrop for larger regional management in light of both the disease and implications of climate change (Linder 2000, Bouriaud et al. 2004, Stephanson and Coe 2017).

The chosen study site is likely in the aftermath stage of BBD, as a majority of the trees are now infected while some appear to remain resistant or are free of BBD causal agents (Shigo 1972, Houston 1983). The land use associated with LDNP and the logging in the 1980s likely influenced the stand level of BBD infection (Hane 2003). Some ecologists have predicted that forests considered in the aftermath stage are not safe from re-infestation as the impacts of climate change and selected microevolution on the scale insect level could synchronize and continue to wreak havoc.

The primary limitation of this study is its scale both geographically and genetically. This is a necessary tradeoff when seeking to intensely monitor and assess a forest stand over several years. There is certainly some question about the degree to which these findings can be extended to other locations. Yet such fine-scale studies are needed to understand how environmental conditions relate to BBD resistance (Evans et al. 2005; Cale et al., 2017). By analyzing a long-term data set on a small scale, we were able to control for the temporal dynamics in disease spread and pressure and thereby detect and estimate the impacts of environmental variables. We found significant evidence that both slope and aspect influence tree susceptibility, a result that is not likely to be limited to only this site by genetic or geographical factors.

Our multifactor analysis also shows the relative impact of environmental factors versus disease spread over space and time as can be seen in Fig. 4. The impact of slope and aspect is approximately as large as the effect of age as measured by DBH, and both are smaller than the effect of being near other diseased trees. Such dynamics are also likely to apply to other disease settings.

5. Conclusions

Beech bark disease continues to go unchecked, and resistance is relatively low throughout the beech range with some areas showing infection rates as high as 99%. Large populations of our beech forests may be in fact non-resistant genetic clones. In cases of both genetic and environmental homogeneity that favor susceptibility, infected trees have a very low survival rate, and, given survival, reinfestation is a possibility. A fifty-year study in the Bartlett Experimental Forest in New Hampshire has prompted Leak (2006) to warn of reinfestation of previously resistant stands with newly evolving disease-complex pathogens. Our study, examining the potential impacts of elevation, slope, aspect, and curvature, acknowledges that recovery efforts are and should continue to be focused on genetic crosses and/or screening to increase the likelihood of survival of beech with resistant stock. However, by adding the environmental attributes into the larger restoration picture, we may be able to more effectively begin to use the broader ecological system itself to increase disease resistance by replanting genetically selected trees within the forest in ways that take advantage of environmental resilience. Further, we also need to continue to manage our forests under the immediate and predicted impacts of climate change.

CRedit authorship contribution statement

Obadiah Mulder: Data curation, Formal analysis, Methodology, Resources, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Robin Sleith:** Conceptualization, Investigation, Methodology. **Kenneth Mulder:** Formal analysis, Methodology, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Natalie Ribarik Coe:** Conceptualization, Data curation, Investigation, Methodology, Project administration, Resources, Supervision, Visualization, Writing - original draft, Writing - review & editing.

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